A3-PS015-TC344

Explainable Home and Last-Mile Connectivity Diagnosis

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Problem Description

Abstract

Internet users often experience slow or unstable connectivity even when a speed test shows acceptable download speed. Existing speed tests do not always explain the reason for the problem.

LastMile Clinician is a software-based diagnostic system that collects network measurements, checks possible faults and explains its findings using transparent decision rules. It aims to identify likely connectivity problems and generate an evidence-based report when the cause cannot be isolated.

Problem description

- Speed alone does not explain overall connection quality.
- Packet loss, latency, DNS failures and congestion can affect applications.
- Users may not know whether a problem is local or beyond their gateway.
- Diagnostic results should be understandable and supported by evidence.

Background

Why is this project needed?

Internet quality depends on more than download speed. Real-time applications such as video calls, online gaming and remote communication are affected by latency, packet loss and jitter.

Key concepts

- Latency: Time taken for data to travel to a destination and back.
- Packet loss: Data packets that fail to reach their destination.
- DNS delay: Time taken to resolve a domain name into an IP address.
- Congestion: Increased delay or loss when network resources are heavily used

Our approach: Use multiple measurements together instead of relying on a single speed-test result.

Literature & RFC Review

Reference framework: RFC 2330

RFC 2330 describes a framework for IP performance metrics and provides a basis for consistent network measurement.

Relevant concepts

- Define what is being measured.
- Use consistent measurement methods.
- Record measurement conditions.
- Interpret results within their limitations.

Gap addressed by our proposed system

Measurements alone may not tell a non-technical user what to do next. LastMile Clinician adds guided diagnostic steps, transparent decision rules and an evidence-based report.

Proposed Solution

LastMile Clinician: Evidence-Guided Network Diagnosis

The proposed system has four main components:

1. **Measurement module** — collects gateway reachability, DNS response, latency and packet-loss measurements.
2. **Diagnostic engine** — evaluates the measurements using readable decision rules.
3. **Guided test planner** — selects the next useful test based on the evidence.
4. **Report generator** — presents the likely cause, supporting evidence, limitations and recommended next steps.

Expected output:

- Likely fault category
- Supporting measurements
- Diagnostic reasoning
- “Inconclusive” when the available evidence is insufficient

Optimization Proposed

Existing approach	Proposed improvement
Focuses mainly on speed	Combines multiple network measurements
Runs tests without guided reasoning	Selects tests based on previous findings
Gives numbers that users must interpret	Explains findings in simple language
May encourage a premature conclusion	Uses evidence checks before a verdict
Provides limited diagnostic history	Generates a structured report

Optimization objective: Reduce unnecessary tests while maintaining diagnostic reliability. These are proposed improvements, not measured results yet. Add experimental comparisons after you run the tests.

Solution Architecture & Design

User / Laptop



Network Measurement Module

Gateway · DNS · Latency · Packet loss



Diagnostic Engine

Rules · Test planner · Evidence checks



Verdict and Report Generator

Likely cause · Evidence · Uncertainty



Results Dashboard

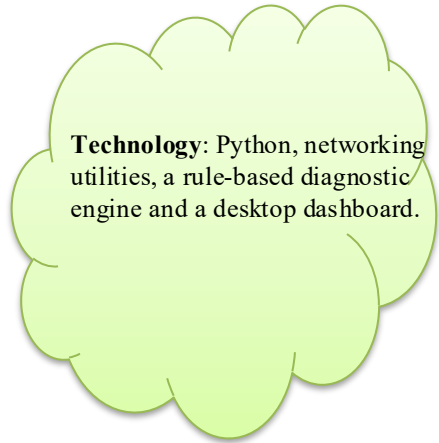


Figure: System Architecture UML Diagram

Data Flow Diagram

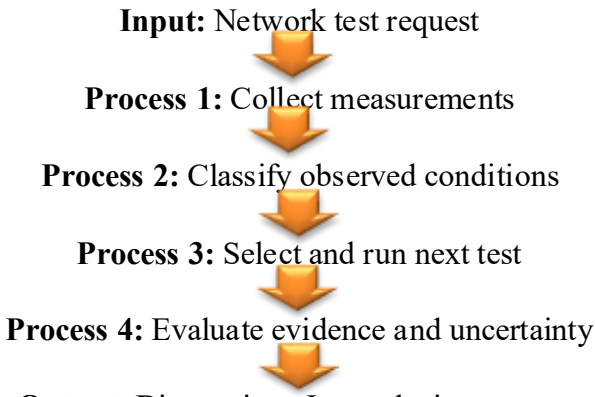


Figure 2: Data Flow Diagram (DFD)

Deployment Timeline

Phase	Work
Phase 1	Study PS-015 and finalize requirements
Phase 2	Build the Python measurement module
Phase 3	Implement decision rules and guided testing
Phase 4	Create controlled fault scenarios
Phase 5	Build the results dashboard and report generator
Phase 6	Test all four faults and evaluate results

Four required test scenarios

- Gateway unreachable
- DNS failure
- Induced packet loss
- Network congestion

Bibliography I

- RFC 2330, Framework for IP Performance Metrics. RFC Editor. https://www.rfc-editor.org/rfc/rfc2330.html

- **Reference:** RFC 2330, Framework for IP Performance Metrics.

Thank You for Your Attention!